AI-Assisted Requirements Candidates – Clinic Appointment System

📋 Problem Summary

Current State Challenges:
The clinic's manual appointment system has created significant operational and compliance risks that require immediate attention.

▼ 🔴 Critical Issues Identified

- **Scheduling Conflicts:** Double bookings occur due to spreadsheet-based coordination
- **Patient No-Shows:** Lack of automated reminders leads to missed appointments
- **Provider Scheduling:** No systematic enforcement of required buffer time between appointments

- **Compliance Gaps:** Absence of audit trails for appointment activities
- **Data Security:** Risk of sensitive medical information exposure through informal communication channels

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✅ Requirements Candidates

The following requirements have been derived from stakeholder elicitation sessions and are presented for validation and prioritization.

Req ID	Requirement Description	Type	Stakeholder Source
`REQ-001`	The system shall prevent double booking of appointment slots for the same provider and time period.	Functional	Clinic Manager
`REQ-002`	The system shall enforce configurable buffer time between consecutive appointments for each provider.	Functional	Medical Staff
`REQ-003`	The system shall send automated appointment reminders to patients via SMS and/or email based on patient preference.	Functional	Patients, Front Desk
`REQ-004`	The system shall maintain a comprehensive audit log of all appointment-related activities including creation, modification, cancellation, and access.	Non-Functional	IT Department, Compliance
`REQ-005`	The system shall NOT include any protected health information (PHI) or medical details in SMS or email communications.	Constraint	Compliance, IT Department
`REQ-006`	The system shall provide real-time availability view for appointment scheduling to prevent conflicts.	Functional	Front Desk, Clinic Manager
`REQ-007`	The system shall allow patients to confirm, reschedule, or cancel	Functional	Patients

	appointments through reminder notifications.		
`REQ-008`	The system shall comply with HIPAA security and privacy requirements for handling patient information.	Constraint	Compliance, IT Department
`REQ-009`	The system shall achieve 99.5% uptime during clinic operating hours (8 AM - 6 PM).	Non-Functional	IT Department, Clinic Manager
`REQ-010`	The system shall generate reports on appointment metrics including no-show rates, cancellations, and provider utilization.	Functional	Clinic Manager

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🎯 Interactive Requirements Review

▼ 📊 Functional Requirements (REQ-001, REQ-002, REQ-003, REQ-006, REQ-007, REQ-010)

These requirements define *what* the system must do. They describe specific behaviors, features, and capabilities.

Review Questions:

- Are these requirements testable and measurable?
  Do they address the core business needs?
  Are there any missing functional capabilities?

▼ ⚡ Non-Functional Requirements (REQ-004, REQ-009)

These requirements define *how well* the system must perform. They specify quality attributes, performance criteria, and operational characteristics.

Review Questions:

- Are the performance metrics realistic and achievable?
  Have we defined all necessary quality attributes?
  Do these align with organizational standards?

▼ 🚧 Constraints (REQ-005, REQ-008)

These are non-negotiable limitations or boundaries within which the solution must operate, often driven by regulatory, technical, or business factors.

Review Questions:

- Are these constraints clearly understood by all stakeholders?
  What is the impact if these constraints are not met?
  Are there additional regulatory or policy constraints to consider?

📝 Next Steps

☐ **Stakeholder Review:** Circulate requirements candidates to all stakeholders for validation

☐ **Prioritization Session:** Conduct MoSCoW or similar prioritization exercise

☐ **Traceability Matrix:** Link requirements to business objectives and elicitation sources

☐ **Acceptance Criteria:** Define detailed acceptance criteria for each requirement

☐ **Risk Assessment:** Identify risks associated with each requirement

☐ **Estimation Session:** Work with technical team to estimate effort and complexity

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💡 **Business Analyst Notes:**
These requirements candidates represent the initial structured output from elicitation activities. They should be considered draft until validated through formal review sessions with all stakeholder groups. Pay particular attention to the constraints (REQ-005, REQ-008) as these are non-negotiable and will significantly impact solution design.